Chapter 7 : Inference for distributions

18 June 2024 11:09

⇒ In this chapter, will meet confidence intervals and significance tests for inference about a population mean, and difference between population means $\mu_1 - \mu_2$.

↳ We no longer assume that population standard deviations are known.

↳ The resulting t-procedures for inference about mean are among the most commonly used statistical methods.

- Inference for the mean of a population:

⇒ When σ is not known, we must estimate σ even though we're primarily interested in μ.

↳ Will use the sample standard deviation s to estimate the population standard deviation σ.

- t-distributions :

⇒ When σ is not known, we estimate it using sample standard deviation S, and then we estimate the standard deviation of $\bar{x}$ by $S/\sqrt{n}$. (Instead of $\sigma/\sqrt{n}$)

↳ This quantity is called the standard error of the sample mean $\bar{x}$ and we denote it by SE.

Standard error :

⇒ When the standard deviation of a statistic (ex: mean of a sampling distribution) is estimated from the data, the result is called the standard error of the statistic. The standard error of the sample mean is

$$SE = \frac{s}{\sqrt{n}}$$

⇒ The standardized sample mean, or one sample z statistic

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

is the basis for z procedures for inference about μ when σ is known. This statistic (z) has a standard Normal Distribution $N(0,1)$.

When we substitute the standard error $s/\sqrt{n}$ for the standard deviation $\sigma/\sqrt{n}$ of $\bar{x}$, the statistic does not have a Normal distribution.

↳ It has a distribution that is new to us called t-distribution.

The t-distribution :

⇒ Suppose that an SRS of size n is drawn from an $N(\mu, \sigma)$ population. Then the one-sample t statistic

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

- has t distribution with n-1 degrees of freedom.

⇒ A particular t distribution is specified by its degree of freedom.

→ The degree of freedom for this t-statistic come from the standard deviation s in the denominator of t.

→ We saw that s has $n-1$ degrees of freedom.

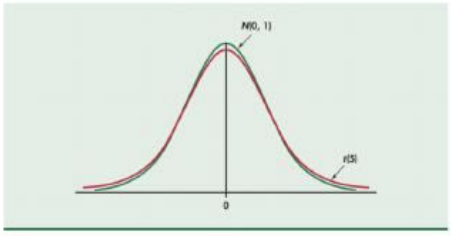
→ This means that there is a different t-distribution for each sample size. (Since t-distribution is dependant on degrees of freedom, in this case its $n-1$)

- There are other t statistics with different degree of freedom, some of which we'll meet later in this chapter.

→ We use $t(k)$ to stand for the t distribution with k degrees of freedom.

→ The density curves of the $t(k)$ distributions are similar shape to the Normal curve. i.e they are symmetric about 0 & are bell-shaped.

FIGURE 7.1 Density curves for the standard Normal and $t(5)$ distributions. Both are symmetric with center 0. The t distribution has more probability in the tails than the standard Normal distribution.



⇒ The above figures compares the density curves of standard Normal & the t-distribution with 5 degrees of freedom.

- The similarity in the shape is apparent, as is the fact that the t distributions have more probability in tails than in center.

→ The greater spread is due to extra variability caused by substituting the random variable s for the fixed parameter σ.

⇒ As the degrees of freedom k increases, the $t(k)$ density curve gets closer to the $N(0,1)$ curve.

→ This reflects the fact that s approaches σ as the sample sizes increases.

- The one sample t-confidence interval :-

⇒ Suppose that an SRS of size n is drawn from a (normal) Population having unknown mean μ. A level C confidence

interval for μ is

$$\bar{x} \pm m$$

⇒ In this formula the margin of error is

$$m = t^* SE. = t^* \frac{s}{\sqrt{n}} \quad \left(\text{as compared to } z^* \frac{\sigma}{\sqrt{n}}\right)$$

⇒ where t^* is the value for the $t(n-1)$ density curve with area C between $-t^*$ and t^*.

↳ This interval is exact when the population distribution is Normal & is approximately correct for large n in other cases.

CASE 7.1

Monthly Household Telephone Expenditures *Trends in Telephone Service* is published by the Industry Analysis and Technology Division of the Federal Communications Commission (FCC). The annual publication contains summary information from a variety of published reports and is designed to answer frequently asked questions about the telephone industry. One section of the report summarizes monthly household expenditures and household personal consumption expenditures for telephone services. The year 2007 marked the first time monthly personal consumption expenditures for cellular service were larger than personal consumption expenditures for landline service.[1] You're interested in how much, on average, a household in your community spends on landline service.

EXAMPLE 7.1 Estimating Monthly Expenditures for Landline Telephone Service

The following data are the monthly dollar amounts for landline telephone service for a random sample of 8 households in your community:

43 47 51 36 50 42 38 41

⇒ We want to find 95% confidence interval for μ.

∴ The sample mean is

$$\bar{x} = \frac{43+47+\ldots+41}{8} = 43.5$$

and the (sample) standard deviation is

$$s = \sqrt{\frac{(43-43.5)^2+(47-43.5)^2+\ldots+(41-43.5)^2}{8-1}} = 5.42.$$

with degrees of freedom $n-1=7$. The standard error of $\bar{x}$ is

$$SE = \frac{s}{\sqrt{n}} = \frac{5.42}{\sqrt{8}} = 1.92$$

from the $C = 95\%$ we find $t^* = 2.365$ for degree of freedom $(df) = 7$.

∴

	$df = 7$	
t^*	1.895	2.365
C	90%	95%

∴ The margin of error,

$$m = t^*(SE) = (2.365)(1.92) = 4.5$$

⇒ The 95% confidence interval is

$$\bar{x} \pm m = 43.5 \pm 4.5$$

$= (39.0, 48.0)$

⇒ So, we're 95% confident that average monthly expenditure for Indian telephone service is between $39 & $48.

⇒ The use of t-confidence interval in example 7.1 rests on two conditions that appear reasonable here.

1) We assume that our random sample is an SRS.

2) We assume the distribution of monthly expenditures is normal.

→ With only 8 observations this cannot be checked, in fact some households may only use cellular telephone service, hence this distribution can be irregular.

→ However, the stemplots of these data has no outliers

5	0 1
4	1 2 3 7
3	5 6

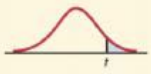
- The one sample t-test :-

⇒ Suppose that an SRS of size n is drawn from a population having unknown mean μ. To test the hypothesis $H_0: \mu = \mu_0$ based on SRS of size n, compute the one-sample t statistic

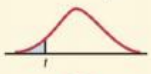
$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

In terms of a random variable T having the $t(n-1)$ distribution, the P-value for a test of H_0 against

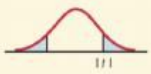
$H_a: \mu > \mu_0$ is $P(T \geq t)$



$H_a: \mu < \mu_0$ is $P(T \leq t)$



$H_a: \mu \neq \mu_0$ is $2P(T \geq |t|)$



These P-values are exact if the population distribution is Normal and are approximately correct for large n in other cases.

EXAMPLE 7.2 Do Average Monthly Landline Expenditures in Your Community Differ from the U.S. Average?

CASE 7.1 Suppose that the overall U.S. average monthly expenditure for landline service is $49. We can test a null hypothesis that the average monthly landline expenditure in your community is equal to this amount. First, we state our hypotheses:

DATA FILE TELEPHONE

$H_0: \mu = 49$

$H_a: \mu \neq 49$

⇒ Recall that $n = 8$, $\bar{x} = 43.5$ and $s = 5.42$.

$$\therefore t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{43.5 - 49}{5.42/\sqrt{8}} = -2.87$$

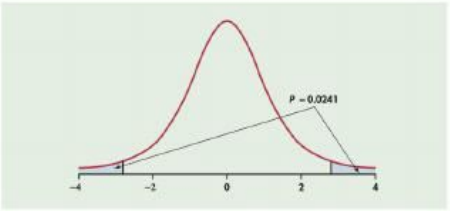
df = 7		
p	0.02	0.01

Because the degrees of freedom are $n - 1 = 7$, this t statistic has the $t(7)$ distribution. Figure 7.2 shows that the P-value is $2P(T \geq 2.87)$, where T has the $t(7)$ distribution. From Table D we see that $P(T > 2.517) = 0.02$ and $P(T > 2.998) = 0.01$. Therefore, we conclude that the P-value

t^*	2.517	2.998

is between $2 \times 0.01 = 0.02$ and $2 \times 0.02 = 0.04$. Software gives the exact value as $P = 0.0241$. Clearly, these data are incompatible with an average of $\mu = \$49$.

FIGURE 7.2 Sketch of the *P*-value calculation for Example 7.2.



EXAMPLE 7.3 One-sided Test for Monthly Landline Expenditures

CASE 7.1

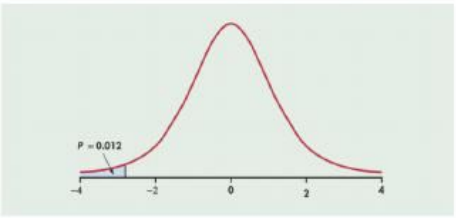


To test whether the average monthly landline expenditure is less than the U.S. average (perhaps expected because your community's cost of living is below the average) our hypotheses are

$$H_0: \mu = 49$$
$$H_a: \mu < 49$$

The t test statistic does not change: $t = -2.87$. As Figure 7.3 illustrates, however, the P-value is now $P(T \leq -2.87)$, half of the value in Example 7.2. From Table D we can determine that $0.01 < P < 0.02$; software gives the exact value as $P = 0.0120$. We conclude that the average monthly expenditure for landline service in your community is less than the U.S. average.

FIGURE 7.3 Sketch of the *P*-value calculation for Example 7.3.



EXAMPLE 7.4 Diversify or Be Sued



An investor with a stock portfolio worth several hundred thousand dollars sued his broker and brokerage firm because lack of diversification in his portfolio led to poor performance. The conflict was settled by an arbitration panel that gave "substantial damages" to the investor.[2] Table 7.1 gives the rates of return for the 39 months that the account was managed by the broker. The arbitration panel compared these returns with the average of the Standard & Poor's 500-stock index for the same period.

TABLE 7.1 Monthly rates of return on a portfolio (percent)

−8.36	1.63	−2.27	−2.93	−2.70	−2.93	−9.14	−2.64
6.82	−2.35	−3.58	6.13	7.00	−15.25	−8.66	−1.03
−9.16	−1.25	−1.22	−10.27	−5.11	−0.80	−1.44	1.28
−0.65	4.34	12.22	−7.21	−0.09	7.34	5.04	−7.24
−2.14	−1.01	−1.41	12.03	−2.56	4.33	2.35	

Consider the 39 monthly returns as a random sample from the population of monthly returns that the brokerage would generate if it managed the account forever. Are these returns compatible with a population mean of $\mu = 0.95\%$, the S&P 500 average? Our hypotheses are

$$H_0: \mu = 0.95$$
$$H_a: \mu \neq 0.95$$

Minitab and SPSS outputs appear in Figure 7.4. Output from other software will look similar.

Here is one way to report the conclusion: the mean monthly return on investment for this client's account was $\bar{x} = -1.1\%$. This differs significantly from the performance of the S&P 500 for the same period ($t = -2.14$, df = 38, $P = 0.039$).

⇒ So the hypothesis test in example 7.4 leads us to conclude that the mean return on the client account differs from that of the stock index.

⇒ Now lets asses the return on the clients account with a confidence interval

□ Example 7.5: Estimating mean monthly returns:

⇒ The mean monthly return on a client portfolio was $\bar{x} = 1.1\%$ & the standard deviation was $s = 5.99\%$.

FIGURE 7.4 Minitab and SPSS outputs for Examples 7.4 and 7.5.

Minitab



SPSS

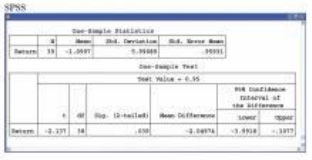
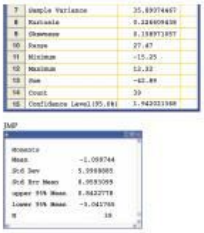


FIGURE 7.5 Excel and JMP outputs for Example 7.5.

Excel





7	Sample Variance	25.69074497
8	Kurtosis	0.224009438
9	Skewness	0.158971957
10	Range	27.47
11	Minimum	-15.25
12	Maximum	12.22
13	Sum	-42.89
14	Count	39
15	Confidence Level(95.0%)	1.942031368

JMP

Moments	
Mean	-1.099744
Std Dev	5.0988085
Std Err Mean	0.9533095
upper 95% Mean	0.8422778
Lower 95% Mean	-3.041766
N	28

DIVERSIFY

JMP outputs for a 95% confidence interval for the population mean μ. Note that Excel gives the margin of error next to the label "Confidence Level (95.0%)" rather than the actual confidence interval. We see that the 95% confidence interval is $(-3.04,\ 0.84)$, or (from Excel) -1.0997 ± 1.9420.

⇒ Because the S&P returns, 0.95% falls outside this interval, we know that μ differs significantly (P value less than 5%) for the 0.95% at the $\alpha = 0.05$ level.

↳ Example 7.4 gave the actual P-value as $P = 0.039$.

⇒ We're not interested in the actual mean, but the difference between the broker's process & the diversified S&P 500 index.

EXAMPLE 7.6 Estimating Difference from a Standard

Following the analysis accepted by the arbitration panel, we are considering the S&P 500 monthly average return as a constant standard. (It is easy to envision scenarios where we would want to treat this type of quantity as random.) The difference between the mean of the investor's account and the S&P 500 is $\bar{x} - \mu = -1.10 - 0.95 = -2.05\%$. In Example 7.5 we found that the 95% confidence interval for the investor's account was $(-3.04,\ 0.84)$. To obtain the corresponding interval for the difference, subtract 0.95 from each of the endpoints. The resulting interval is $(-3.04 - 0.95,\ 0.84 - 0.95)$, or $(-3.99,\ -0.11)$. We conclude with 95% confidence that the underperformance will be between −3.99% and −0.11%. This estimate helps to set the compensation owed to the investor.

DIVERSIFY

◦ Matched pairs t procedures :—

⇒ One common comparitive design makes use of single-sample procedures.

↳ In a matched pairs study, subjects are matched in pairs and the outcomes are compared against each matched pairs.

↳ For example, an experiment to compare two advertising campaigns might use pairs of subject that are the same age, sex, or income level.

↳ The experimenter can toss a coin (random choice) to assign the two campaigns to the two subjects in each pairs.

⇒ One situations is before & after observations on the same subjects, as illustrated in the next example.

EXAMPLE 7.7 The Effects of Language Instruction

Similar to Model comparision

A company contracts with a language institute to provide individualized instruction in foreign languages for its executives who will be posted overseas. Is the instruction effective?

Last year, 20 executives studied French. All had some knowledge of French, so they were given the Modern Language Association's listening test of understanding of spoken French before the instruction began. After several weeks of immersion in French, the executives took the listening test again. (The actual French spoken in the two tests was different, so that simply taking the first test should not improve the score on the second test.)

Table 7.2 gives the pretest and posttest scores. The maximum possible score on the test is 36.[3] To analyze these data, subtract the pretest score from the posttest score to obtain the improvement for each executive. These 20 differences form a single sample. They appear in the "Gain" columns in Table 7.2. The first executive, for example, improved from 32 to 34, so the gain is $34 - 32 = 2$.

FRENCH

TABLE 7.2 French listening scores for executives

Executive	Pretest	Posttest	Gain	Executive	Pretest	Posttest	Gain
1	32	34	2	11	30	36	6
2	31	31	0	12	20	26	6

3	29	35	6	13	24	27	3
4	10	16	6	14	24	24	0
5	30	33	3	15	31	32	1
6	33	36	3	16	30	31	1
7	22	24	2	17	15	15	0
8	25	28	3	18	32	34	2
9	32	26	−6	19	23	26	3
10	20	26	6	20	23	26	3

To assess whether the institute significantly improved the executives' comprehension of spoken French, we test

$H_0: \mu = 0$ → exactly same

$H_a: \mu > 0$

Here μ is the mean improvement that would be achieved if the entire population of executives received similar instruction. The null hypothesis says that no improvement occurs, and H_a says that positive scores are higher on the average.

The 20 differences have

$$\bar{x} = 2.5 \text{ and } s = 2.893$$

The one-sample t statistic is therefore

$$t = \frac{\bar{x} - 0}{s/\sqrt{n}} = \frac{2.5}{2.893/\sqrt{20}} = 3.86$$

⇒ The table

$df = 19$

p	0.001	0.0005
t^*	3.579	3.883

⇒ The table shows that the $t = 3.86$ lies between upper 0.001 & 0.0005 critical values of the $t(19)$ distribution

→ Software gives the value $P = 0.00053$.

→ We have strong evidence that post-test score was significantly higher (since null hypothesis is incompatible).

→ The test would be reported in the form

↳ The improvement was significant ($t = 3.86$, $df = 19$, $P = 0.00053$)".

⇒ The significant test showed that the mean scores improved.

→ A critic would argue that the test does not show that the instruction caused the improvement.

→ Maybe the executives where less nervous taking the test a second time.

the executives might simply be less nervous when taking the test a second time. A more elaborate study with a control group who receive no instruction would be more convincing, but the company is unlikely to assign executives to "no instruction." Also note that the subjects all knew that they would receive overseas assignments. Our conclusions apply only to the population of such executives, not to the larger population of all the company's managers.

⇒ A statistically significant, but very small improvement in language ability would not justify the expense of the individualized instruction.

→ A confidence interval allows us to estimate the amount of the improvement.

EXAMPLE 7.8 The Effects of Language Instruction

Let's compute a 90% confidence interval for the mean improvement in the entire population. The standard error is

$$\text{SE} = \frac{s}{\sqrt{n}} = \frac{2.893}{\sqrt{20}} = 0.6469$$

The margin of error is

$$m = t^* \text{SE} = (1.729)(0.6469) = 1.12$$

where the critical value $t^* = 1.729$ comes from Table D. The confidence interval is

$$\bar{x} \pm m = 2.5 \pm 1.12$$
$$= (1.38,\ 3.62)$$

~~The estimated average improvement is 2.5 points, with margin of error 1.12 for 90% confidence. Though statistically significant, the effect of the instruction was rather small.~~

~~A look at the data discloses one reason for the small average improvement. Several of the executives had pretest scores close to the maximum of 36. They could not improve their scores very much even if their mastery of French increased substantially. This is a weakness in the listening test that is the measuring instrument in this study.~~

⇒ Here are some points to remember concerning matched pairs:

- A matched pairs analysis is needed when subjects or experimental units are matched in pairs or there are two measurements or observations on each individual or experimental unit and our question of interest concerns the difference between the two measurements. Frequently, the observations are "before" and "after" measures.
- The analysis is based on the *difference* between the two measures for each individual or pair of experimental units.
- Use the one-sample confidence interval and significance-testing procedures that we learned in this section.

o Robustness of the t-Procedures :-

⇒ The matched pairs t procedure produces one-sample t confidence intervals & significance test for the differences.

→ They are therefore based on the assumption that the population differences has a Normal distribution.

⇒ The result of one-sample t-test procedure are exactly correct only when the population is Normal.

→ Real populations are never Normal. All inference procedures are based on some conditions, such as Normality.

→ Procedures that are not strongly affected by violation of condition are called robust

Robust Procedures

A statistical inference procedure is called **robust** if the probability calculations required are insensitive to violations of the conditions that usually justify the procedure.

⇒ The condition that the population be normal rules out outliers, so the presence of outliers shows this condition is not fulfilled.

→ The t-procedures are not robust against outliers, because $\bar{x}$ & s are resistant to outliers.

⇒ Fortunately, the t-procedures are quite robust against non-Normal population.

→ Except when outliers or strong skewness is present, particularly when the sample size is large.

⇒ The t procedures rely only on the Normality of the sample mean $\bar{x}$.

→ This condition is satisfied when the population is normal.

→ The central limit theorem states that the mean $\bar{x}$ from a

large sample follows a ND closely even when the individual observations are not normally distributed.

⇒ Before using the t-procedures for small samples, make a normal quantile plot, stemplot, or box plot to check for skewness & outliers.

⇒ For most purposes, the one sample t-procedures can be safely used when $n \geq 15$ unless an outlier or clearly marked skewness is present.

- *Sample size less than 15:* Use t procedures if the data are close to Normal. If the data are clearly non-Normal or if outliers are present, do not use t.
- *Sample size at least 15:* The t procedures can be used except in the presence of outliers or strong skewness.
- *Large samples:* The t procedures can be used even for clearly skewed distributions when the sample is large, roughly $n \geq 40$.

The French instruction data are a borderline case: there are 20 observations, with one clear outlier. The t procedures give approximately correct results, but we would be happier if a good reason, such as illness on the posttest date, justified removing the outlier.

- The power of the t test :

⇒ To calculate the power, we assume a fixed level significance, usually $\alpha = 0.05$.

⇒ Calculation of exact power of t test takes in account the estimation of σ by s & requires special software.

↳ An approximate calculation that is based on assuming that σ is known is generally adequate for planning a study.

↳ This calculation is very much like a z-test.

1) Write the event, in terms of $\bar{x}$, that the test rejects H_0.

2) Find the probability of this event when the population mean has the alternative value.

EXAMPLE 7.9 Is the Sample Size Large Enough?

The company in Example 7.7 is planning to evaluate the next language instruction session. They decide that an improvement of 2 points or more is large enough to be useful in practice. Can they rely on a sample of 20 executives to detect an improvement of this size?

They wish to compute the power of the t test for

$$H_0: \mu = 0$$
$$H_a: \mu > 0$$

against the alternative $\mu = 2$ when $n = 20$. This gives us most of the information we need to compute the power. The other important piece is a rough guess of the size of σ. In planning a large study, a pilot study is often run for this and other purposes. In this case, listening-score improvements in past language instruction programs for similar executives have had sample standard deviations of about 3. We will therefore take both $\sigma = 3$ and $s = 3$ in our approximate calculation.

Step 1. *The t test with 20 observations rejects H_0 at the 5% significance level if the t statistic*

$$t = \frac{\bar{x} - 0}{s/\sqrt{20}}$$

exceeds the upper 5% point of $t(19)$, which is 1.729. Taking $s = 3$, the event that the test rejects H_0 is therefore

$$t = \frac{\bar{x}}{3/\sqrt{20}} \geq 1.729$$

$$\bar{x} \geq 1.729\frac{3}{\sqrt{20}}$$

$$\bar{x} \geq 1.160$$

Step 2. *The power is the probability that $\bar{x} \geq 1.160$ when $\mu = 2$. Taking $\sigma = 3$, we find this probability by standardizing $\bar{x}$:*

$$P(\bar{x} \geq 1.160 \text{ when } \mu = 2) = P\left(\frac{\bar{x} - 2}{3/\sqrt{20}} \geq \frac{1.160 - 2}{3/\sqrt{20}}\right)$$

$$= P(Z \geq -1.252)$$
$$= 1 - 0.1056 = 0.8944$$

A true difference of 2 points in the population mean scores will produce significance at the 5% level in 89% of all possible samples. The company can be reasonably confident of detecting a difference this large.

- Inference for Non-Normal Populations :-

⇒ Three general strategies are available :-

1) In some cases a distribution other than Normal distribution describes the data well. There are many non-Normal models for data, and inference procedures for these models are available.

2) Because skewness is the chief departure to the use of t-procedures on data without outliers, you can attempt to transform skewed data so that the distribution is symmetric, and as close to Normal as possible.
   ↳ Confidence intervals & P-values from the t-procedures will be quite accurate for even moderately sized samples. Methods are generally available for transforming the results back to the original scale.

3) The third strategy is to use distribution-free procedures. Such procedures do not assume that the population distribution has specific form, such as Normal.
   ↳ Distribution-free procedures are often called non-parametric procedures. ex: The bootstrap.

⇒ If the data is really close to Normal, distribution free tests have less power than t-tests.
   ↳ They also don't answer the same question.
      ↳ The t test concerns population mean. Distribution-free tests often about population median, as is natural for distributions that may be skewed.

- The sign test :

⇒ The simplest distribution-free test, and one of the most useful, is the sign test.
   ↳ The test gets its name from the fact that we only look at the sign of the differences, not their actual values.

⇒ The following examples illustrates that :

EXAMPLE 7.10 The Effects of Language Instruction

Example 7.7 (page 405) gives data on the improvement in French listening scores after individualized instruction. In that example we used the matched pairs t test on these data, despite granularity and an outlier, which make the P-value only roughly correct. The sign test is based on the following simple observation: of the 17 executives whose scores changed, 16 improved and only 1 did more poorly. This is evidence that the instruction improved French listening skills.

To perform a significance test based on the count of executives whose scores improved, let p be the probability that a randomly chosen executive would improve if she attended the institute. The null hypothesis of "no effect" says that the posttest is just a repeat of the pretest with no change in ability, so an executive is equally likely to be better or either test. We therefore want to test

$$H_0: p = 1/2$$

equally likely to perform good in both test.

7.1 Inference for the Mean of a Population 411

The 17 executives whose scores changed are 17 independent trials, so the number who improve has the binomial distribution $B(17, 1/2)$ if H_0 is true. The P-value for the observed count 16 is therefore $P(X \geq 16)$, where X has the $B(17, 1/2)$ distribution. You can compute this probability with software or from the binomial probability formula (page 309):

$$P(X \geq 16) = P(X = 16) + P(X = 17)$$
$$= \binom{17}{16}\left(\frac{1}{2}\right)^{16}\left(\frac{1}{2}\right)^{1} + \binom{17}{17}\left(\frac{1}{2}\right)^{17}\left(\frac{1}{2}\right)^{0}$$
$$= (17)\left(\frac{1}{2}\right)^{17} + \left(\frac{1}{2}\right)^{17}$$
$$= 0.00014$$

Null hypothesis can be rejected

As in Example 7.7, there is very strong evidence that participation in the institute has improved performance on the listening test.

⇒ There are variety of sign tests, all based on counts, and the binomial distribution.

↳ The sign test for matched pairs is the most useful.

↳ The null hypothesis of the "no effect" is then always $H_0 : p = \frac{1}{2}$.

- Sign test for Matched Pairs :-

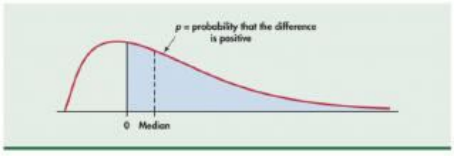
⇒ Ignore pairs with difference 0 (i.e with same performance before & after treatment); the number of trials n is the count of the remaining pairs.

↳ The test statistic is the count of X pairs with a positive difference. P-values for X are based on the binomial $B(n, 1/2)$ distribution.

⇒ The sign test is exact testing if so exact testing the hypothesis that the median of the differences is 0.

↳ If p is the probability that a difference is positive, then $p = 1/2$ when the median is $1/2$.

FIGURE 7.7 Why the sign test tests the median difference: when the median is greater than 0, the probability p of a positive difference is greater than 1/2, and vice versa.



⇒ The sign test of $H_0 : p = 1/2$ against $H_a : p > 1/2$ is a test of

H_0 : Population median = 0

H_a : Population median > 0

- Comparing two Means :-

⇒ How do small businesses that fail differ from those that succeed?

↳ Business school researchers compare two samples of new firms.

↳ One sample of failed businesses.

↳ And one sample of firms still going

⇒ The study compares two random samples, one from each of two different populations.

⇒ Which of the two incentive packages will lead to higher use of credit cards?

⇒ Two sample problems like these are among the most common situations encountered in statistical practice.

◦ Two sample problems:

→ The goal of inference is to compare the response in two groups.

→ Each group is considered to be a sample from a distinct population.

→ The responses in each group are independent of those in other group.

⇒ In two-samples problem, there is no matching of units in the two samples, and the two samples may differ in size.

↳ Inference procedure of two sample differ from the matched problem.

⇒ When both population distributions are symmetric, and especially when they are at least approximately Normal, a comparision of mean responses in the two populations is most often the goal of inference.

Here is the notation that we will use to describe the two populations:

Population	Variable	Mean	Standard deviation
1	x_1	μ_1	σ_1
2	x_2	μ_2	σ_2

⇒ We want to compare the two population means, either by giving a confidence interval for $\mu_1 - \mu_2$, or by testing the hypothesis of

on two independent SRSs, one from each population. Here is the notation that describes the samples:

Population	Sample size	Sample mean	Sample standard deviation
1	n_1	$\bar{x}_1$	s_1
2	n_2	$\bar{x}_2$	s_2

Throughout this section, the subscripts 1 and 2 show the population to which a parameter or a sample statistic refers.

◦ The two-sample z-statistics :—

⇒ The natural estimator of difference $\mu_1 - \mu_2$ is the difference between the sampling means.

↳ If we're to base our inference on this statistic, we must know its sampling distribution

⇒ Because the sampling distributions have same mean ∴ the mean of difference is the difference of means.

↳ ∴ mean of the difference $\bar{x}_1 - \bar{x}_2$ (sampling dist of $\bar{x}_1 - \bar{x}_2$) $= \mu_1 - \mu_2$

⇒ The variance of difference $\bar{x}_1 - \bar{x}_2$ is,

$\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$

⇒ If the two population distribution are both normal, then the distribution of $\bar{x}_1 - \bar{x}_2$ is also normal.

↳ Because each sample mean alone is N.D, and difference of independent N.D is also Normal.

Two-Sample z Statistic

Suppose that $\bar{x}_1$ is the mean of an SRS of size n_1 drawn from an $N(\mu_1, \sigma_1)$ population and that $\bar{x}_2$ is the mean of an independent SRS of size n_2 drawn from an $N(\mu_2, \sigma_2)$ population. Then the **two-sample z statistic**

$$z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

has the standard Normal $N(0, 1)$ sampling distribution.

In the unlikely event that both population standard deviations are known, the two-sample z statistic is the basis for inference about $\mu_1 - \mu_2$. Exact z procedures are seldom used, however, because σ_1 and σ_2 are rarely known. In Chapter 6, we discussed the one-sample z procedures to introduce the ideas of inference. Now we pass immediately to the more useful t procedures.

- The two samples t-procedure:

⇒ In practice the two population standard deviations σ_1 & σ_2 are not known.

↳ We substitute the standard errors for standard deviations in the two-sample z-statistics. The result is the two-sample t-statistics:

Image upload failed. Please try again

Unfortunately, the statistics doesn't have a t-distribution. A t distribution replaces an $N(0,1)$ distribution only when a single standard deviation σ in a z statistics is replaced by s.

↳ Nonetheless, we can approximate the distribution of the two-sample t statistic by using the $t(k)$ distribution with an approximation for the degrees of freedom k.

⇒ There are two procedures used in practice:

1) Use an approximation known as Satterthwaite approximation to calculate a value k from the data. In general k will not be an integer.

2) Use degrees of freedom k equal to the smaller of $n_1 - 1$ & $n_2 - 1$.

↳ This approximation is conservative, because the margin of error for confidence intervals

are a bit larger, so the true confidence level is larger than C.

→ & true P-values are also smaller, so we're less likely to reject H_0.

- The two-sample t-significance test

Two-Sample t Significance Test

Draw an SRS of size n_1 from a Normal population with unknown mean μ_1 and an independent SRS of size n_2 from another Normal population with unknown mean μ_2. To test the hypothesis H_0: $\mu_1 = \mu_2$, compute the **two-sample t statistic**

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

and use P-values or critical values for the $t(k)$ distribution, where the degrees of freedom k are either approximated by software or are the smaller of $n_1 - 1$ and $n_2 - 1$.

EXAMPLE 7.11 Wine Labels with Animals?

Traditional brand research argues that successful logos are ones that are highly relevant to the product they represent. For example, pictures of grapes or a vineyard would be considered highly appropriate for a table wine. However, a market research firm recently reported that nearly 20% of all table wine brands introduced in the last three years feature an animal on the label. Since animals have little to do with the product, why are marketers using this tactic?

A recent study sheds some light on this tactic's success.[14] The researchers found that consumers have an easier time processing visual information when they are "primed" – in other words, when they've thought about the image earlier in an unrelated context. To demonstrate this, one of the researchers' experiments compared product preference for a group of consumers who were "primed" and a group of consumers who were not primed (actually, the study involved three groups but we will focus on just these two here). A bottle of MagicCoat pet shampoo, with a picture of a collie on the label, was the product, and participants indicated their attitude toward this product on a seven-point scale (from 1-dislike very much to 7-like very much). Prior to giving this score, however, participants were asked to do a "word find" where four of the words were common across groups (pet, grooming, bottle, label) and four were either related to the image (dog, collie, puppy, woof) or conflicted with the image (cat, feline, kitten, meow). Since their journal article reported only summary information, we generated the following data to approximate their results. We assume that there are 22 subjects in the "primed" group and 20 subjects in the "nonprimed" group.

A back-to-back stemplot of the results suggests a slight skewness in the responses, perhaps more so in the primed group, but has no obvious outliers. Because the responses are integers, the distributions are certainly non-Normal, but given the sample sizes and lack of serious departures from Normality, we will press on using the t procedures.

Nonprimed		Primed
xx	1	
xx	2	xx
xxxxxxxxxxx	3	xxx
xxx	4	xxxxxxxxx
x	5	xxxxxxx
	6	
	7	

In general, the scores of the primed group appear to be higher than those of the nonprimed group. The summary statistics are

Group	n	$\bar{x}$	s
Primed	22	4.00	0.93
Nonprimed	20	2.95	0.94

Because the researchers hope to show that the primed group (Group 1) views the product more favorably than the nonprimed group (Group 2), the hypotheses are

$$H_0: \mu_1 = \mu_2$$
$$H_a: \mu_1 > \mu_2$$

The two-sample t test statistic is

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{4.00 - 2.95}{\sqrt{\frac{0.93^2}{22} + \frac{0.94^2}{20}}} = 3.63$$

The P-value for the one-sided test is $P(T \geq 3.63)$. Software gives the approximate P-value as 0.0004 and uses 39.5 as the degrees of freedom. For the second approximation, the degrees of freedom k are equal to the smaller of

$$n_1 - 1 = 22 - 1 = 21 \text{ and } n_2 - 1 = 20 - 1 = 19$$

	df = 19	
p	0.001	0.0005
t^*	3.579	3.883

Comparing $t = 3.63$ with the entries in Table D for 19 degrees of freedom, we see that P lies between 0.0005 and 0.001. The data strongly suggest that consumers who were primed view the dog shampoo more favorably ($t = 3.63$, df = 19, $P < 0.001$).

- The two-sample t confidence interval :-

⇒ The same ideas that we used for the two-sample t significance tests also apply to two-sample t confidence intervals.

Two-Sample *t* Confidence Interval

Draw an SRS of size n_1 from a Normal population with unknown mean μ_1 and an independent SRS of size n_2 from another Normal population with unknown mean μ_2. The **confidence interval for $\mu_1 - \mu_2$** given by

$$(\bar{x}_1 - \bar{x}_2) \pm t^* \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

[redacted]

has confidence level at least C no matter what the population standard deviations may be. ~~The margin of error is~~

$$t^* \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

Here, t^* is the value for the $t(k)$ density curve with area C between $-t^*$ and t^*. The value of the degrees of freedom k is approximated by software or we use the smaller of $n_1 - 1$ and ~~$n_2 - 1$~~.

Here degree of freedom is used.

⇒ To complete the analysis of the preference scores we examined, we need to describe the size of the treatment effect. We do this with confidence interval for the difference between group means.

EXAMPLE 7.12 How Large an Increase in Preference?

DATA FILE BRANDPREFERENCE

We will find a 95% confidence interval for the mean increase. The interval is

$$(\bar{x}_1 - \bar{x}_2) \pm t^* \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} = (4.00 - 2.95) \pm t^* \sqrt{\frac{0.93^2}{22} + \frac{0.94^2}{20}}$$
$$= 1.05 \pm (t^* \times 0.29)$$

Using software, the degrees of freedom are 39.5 and $t^* = 2.022$. This approximation gives

$$1.05 \pm (2.022 \times 0.29) = 1.05 \pm 0.58 = (0.47,\ 1.63)$$

The conservative approach uses the $t(19)$ distribution. Table D gives $t^* = 2.093$. With this approximation we have

$$1.05 \pm (2.093 \times 0.29) = 1.05 \pm 0.60 = (0.45,\ 1.65)$$

The conservative approach does give a larger interval than the more accurate approximation used by software. However, the difference is very small.

We estimate the mean increase in preference when primed to be about 1 point, but with a margin of error of slightly more than half a point. Although we have good evidence of an increase, the data do not allow a very precise estimate of the size of the average increase.

◦ Robustness of the two-samples procedure:

⇒ The two samples t-procedure are more robust than the one sample t methods.

↳ When the sizes of the two samples are equal & the distributions of the two populations being compared have similar shapes, probability values from the t table are quite accurate for a broad range of distributions when the sample sizes are as small as $n_1 = n_2 = 5$.

↳ When two populations have different shapes, larger samples are needed.

EXAMPLE 7.13 Wheat Prices

The U.S. Department of Agriculture (USDA) uses sample surveys to produce important economic estimates.[16] One pilot study estimated wheat prices in July and in September using independent samples of wheat producers in the two months. Here are the summary statistics, in dollars per bushel:

Month	n	$\bar{x}$	s
September	45	$7.43	$0.24

July	90	$7.16	$0.27

The September prices are higher on the average. But we have data from only a limited number of producers each month. Can we conclude that national average prices in July and September are not the same? Or are these differences merely what we would expect to see due to random variation?

Because we did not specify a direction for the difference before looking at the data, we choose a two-sided alternative. The hypotheses are

$$H_0: \mu_1 = \mu_2$$
$$H_a: \mu_1 \neq \mu_2$$

Because the samples are moderately large, we can confidently use the t procedures even though we lack the detailed data and so cannot verify the Normality condition.

The two-sample t statistic is

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{7.43 - 7.16}{\sqrt{\frac{0.24^2}{45} + \frac{0.27^2}{90}}} = 5.91$$

The conservative approach finds the P-value by comparing 5.91 to critical values for the $t(44)$ distribution because the smaller sample has 45 observations. We must double the table tail area p because the alternative is two-sided.

df = 40	
p	0.0005
t^*	3.551

Table D does not have entries for 44 degrees of freedom. When this happens, we use the next smaller degrees of freedom. Our calculated value of t is larger than the $p = 0.0005$ entry in the table. Doubling 0.0005, we conclude that the P-value is less than 0.001. The data give conclusive evidence that the mean wheat prices were higher in September than they were in July ($t = 5.91$, $df = 44$, $P < 0.001$).

In this and other examples, we can choose which population to label 1 and which to label 2. After inspecting the data, we chose September as Population 1 because this choice makes the t statistic a positive number. This avoids any possible confusion from reporting a negative value for t. Choosing the population labels is *not* the same as choosing a one-sided alternative after looking at the data. Choosing hypotheses after seeing a result in the data is a violation of sound statistical practice.

- Inference for small samples :-

⇒ The power of significance tends to be low, and the margin of error of confidence intervals tends to be large with small samples.

↳ Despite this, we can draw important conclusions from studies with small sample sizes. If the size of an effect is as large as it was in the wheat example, it should be still evident if the ns are small.

EXAMPLE 7.14 More about Wheat Prices

WHEAT

In the setting of Example 7.13, a quick survey collects prices from only 5 producers each month. The data are

Month	Price of wheat ($/bushel)				
September	$7.3325	$7.5575	$7.3125	$7.3600	$7.5550
July	$6.8250	$7.3025	$7.0275	$7.0825	$7.3000

The prices are reported to the nearest quarter of a cent. First, examine the distributions with a back-to-back stemplot after rounding each price to the nearest cent.

September		July
	6.8	3
	6.9	
	7.0	38
	7.1	
	7.2	
631	7.3	00
	7.4	
66	7.5	

The pattern is reasonably clear. Although there is variation among prices within each month, the top 5 prices are all from September and the 5 lowest prices are from July.

A significance test can confirm that the difference between months is too large to easily arise just by chance. We test

$$H_0: \mu_1 = \mu_2$$
$$H_a: \mu_1 \neq \mu_2$$

The price is higher in September ($t = 3.00$, $df = 6.6$, $P = 0.0214$). The difference in sample means is 31.6 cents.

- Satterthwaite approximation for the degrees of freedom

⇒ We noted earlier that the two sample t-statistic doesn't have exact t-distribution, moreover exact distribution changes as the unknown population standard deviations σ_1 & σ_2 change.

⇒ However the distribution can be approximated by a t-distribution with degrees of freedom given by

$$\left(s_1^2 + s_2^2 \right)^2$$

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{1}{n_1-1}\left(\frac{s_1^2}{n_1}\right)^2 + \frac{1}{n_2-1}\left(\frac{s_2^2}{n_2}\right)^2}$$

⭐ The Satterthwaite approximation is quite accurate when both sample sizes n_1 & n_2 are 5 or larger.

↳ The number of degrees of freedom (df) given by the satterthwaite approximation is always at least as large as ⭐ the smaller of n_1-1 & n_2-1.

↳ And the df is never larger than n_1+n_2-2.

- The pooled two-samples t procedures :-

⇒ There is one situation in which a t-statistic for comparing two means has exactly a t distribution.

↳ Suppose that the two N.D have the same standard deviation.
In this case we only substitute only a single standard error in a z-statistic, and the resulting t-statistic has a t distribution.

⇒ Both sample variances s_1^2 and s_2^2 estimate σ^2. The best way to estimate these two estimates is to average them with weights equal to their degrees of freedom.

↳ This gives more weight to the information of larger sample.

The resulting estimator of σ^2 is

$$s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}$$

⇒ This is called pooled estimator of σ^2.

⇒ When both populations have variance σ^2, the addition rule for variance says that $\bar{x}_1 - \bar{x}_2$ has variance equal to the sum of individual variances, which is

$$\frac{\sigma^2}{n_1} + \frac{\sigma^2}{n_2} = \sigma^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right).$$

⇒ The standardized difference in this case will be

$$z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sigma\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}.$$

⭐⭐ This is the special two-sample z-statistics for the case in which the ⭐ populations have same σ.

⭐ Replacing the unknown σ by an estimate s_p gives a t-statistic. The degrees ⭐⭐

of freedom are n_1+n_2-2 , the sum of degrees of freedom of the two sample variances.

↳ because the information is still dependent on x_n & y_n redundant hence n_1+n_2-2.

Pooled Two-Sample t Procedures

Draw an SRS of size n_1 from a Normal population with unknown mean μ_1 and an independent SRS of size n_2 from another Normal population with unknown mean μ_2. Suppose that the two populations have the same unknown standard deviation. A level C confidence interval for $\mu_1 - \mu_2$ is

$$(\bar{x}_1 - \bar{x}_2) \pm t^* s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

Here t^* is the value for the $t(n_1 + n_2 - 2)$ density curve with area C between $-t^*$ and t^*.

To test the hypothesis $H_0: \mu_1 = \mu_2$, compute the **pooled two-sample t statistic**

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

because $H_0: \mu_0 = 0$ ∴ $\frac{\bar{x} - \mu}{\sigma_{std}}$

and use P-values from the $t(n_1 + n_2 - 2)$ distribution.



Chris Hondros/Getty Images

CASE 7.2 DATA FILE COMPANIES

Healthy Companies versus Failed Companies In what ways are companies that fail different from those that continue to do business? To answer this question, one study compared various characteristics of 68 healthy and 33 failed firms.[17] One of the variables was the ratio of current assets to current liabilities. Roughly speaking, this is the amount that the firm is worth divided by what it owes. The data appear in Table 7.3.

TABLE 7.3 Ratio of current assets to current liabilities

Healthy firms						Failed firms		
1.50	0.10	1.76	1.14	1.84	2.21	0.82	0.89	1.31
2.08	1.43	0.68	3.15	1.24	2.03	0.05	0.83	0.90
2.23	2.50	2.02	1.44	1.39	1.64	1.68	0.99	0.62
0.89	0.23	1.20	2.16	1.80	1.87	0.91	0.52	1.45
1.91	1.67	1.87	1.21	2.05	1.06	1.16	1.32	1.17
0.93	2.17	2.61	3.05	1.52	1.93	0.42	0.48	0.93
1.95	2.61	1.11	0.95	0.96	2.25	0.88	1.10	0.23
2.73	1.56	2.73	0.90	2.12	1.42	1.11	0.19	0.13
1.62	1.76	2.22	2.80	1.85	0.96	2.03	0.51	1.12
1.71	1.02	2.50	1.55	1.69	1.64	0.92	0.26	1.15
1.03	1.80	0.67	2.44	2.30	2.21	0.13	0.88	0.09
1.96	1.81							

As usual, we first examine the data. Histograms for the two groups of firms are given in Figure 7.9. Normal curves with mean and standard deviation equal to the sample values are superimposed on the histograms. The distribution for the healthy firms looks more Normal than the distribution for the failed firms. It appears that there may actually be two subgroups of data for the failed firms. This is seen more clearly in the back-to-back stemplot given in Figure 7.10. However, there are no outliers or strong departures from Normality that will prevent us from using the t procedures for these data. Let's compare the mean current assets to current liabilities ratio for the two groups of firms using a significance test.

FIGURE 7.9 Histograms for assets to liabilities ratios, for Example 7.16.

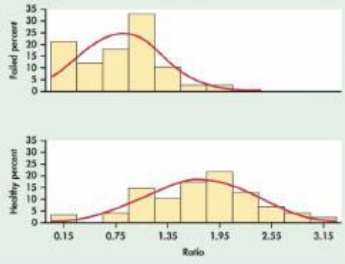


FIGURE 7.10 Back-to-back stemplot for assets to liabilities ratios, for Example 7.16.

Failed firms		Healthy firms
11100	0	1
22	0	2
5544	0	
6	0	66
9999988888	0	8899999
111111	1	00011
33	1	2223
4	1	4445555
6	1	66666777
	1	8888888899999
0	2	00001111
	2	222223
	2	455
	2	6677
	2	8
	3	01

EXAMPLE 7.16 Do Mean Asset/Liability Ratios Differ?

CASE 7.2 DATA FILE COMPANIES

Take Group 1 to be the firms that were healthy and Group 2 to be those that failed. The question of interest is whether or not the mean ratio of current assets to current liabilities is different for the two groups. We therefore test

$$H_0: \mu_1 = \mu_2$$
$$H_a: \mu_1 \neq \mu_2$$

Here are the summary statistics:

Group	Firms	n	$\bar{x}$	s
1	Healthy	68	1.7256	0.6393
2	Failed	33	0.8236	0.4811

The sample standard deviations are fairly close. A difference this large is not particularly unusual even in samples this large. We are willing to assume equal population standard deviations. The pooled sample variance is

$$s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}$$
$$= \frac{(67)(0.6393)^2 + (32)(0.4811)^2}{68+33-2}$$
$$= 0.35141$$

so that

$$s_p = \sqrt{0.35141} = 0.5928$$

The pooled two-sample t statistic is

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}}$$
$$= \frac{1.7256 - 0.8236}{0.5928\sqrt{\frac{1}{68}+\frac{1}{33}}} = 7.17$$

The P-value is $P(T \geq 7.17)$, where T has the $t(99)$ distribution.

In Table D we have entries for 80 and 100 degrees of freedom. We will use the entries for 100. Our calculated value of t is larger than the $p = 0.0005$ entry in the table. Doubling 0.0005, we conclude that the two-sided P-value is less than 0.001. Statistical software gives a similar result—there is no practical need to report the exact P-value when it is smaller than 0.001.

⇒ A p-value is rarely a complete summary of a statistical analysis.

↳ To make a judgement regarding the size of difference we need confidence intervals.

EXAMPLE 7.17 How Different Are Mean Asset/Liability Ratios?

The difference in mean current assets to current liabilities ratios for healthy versus failed firms is

$$\bar{x}_1 - \bar{x}_2 = 1.7256 - 0.8236 = 0.902$$

For a 95% margin of error we will use the critical value $t^* = 1.984$ from the $t(100)$ distribution. The margin of error is

$$t^* s_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}} = (1.984)(0.5928)\sqrt{\frac{1}{68}+\frac{1}{33}}$$
$$= 0.25$$

We report that the successful firms have current assets to current liabilities ratios that average 0.90 higher than failed firms, with margin of error 0.25 for 95% confidence. Alternatively, we are 95% confident that the difference is between 0.65 and 1.15.

The pooled t procedures are therefore a bit risky. They are reasonably robust against both non-Normality and unequal standard deviations when the sample sizes are nearly the same. When the samples are quite different in size, the pooled t procedures become sensitive to unequal standard deviations and should be used with caution unless the samples are large. Unequal standard deviations are quite common. In particular, it is common for the spread of data to increase when the center moves up, as in the data given in Case 7.2. We recommend regular use of the unpooled t procedures, particularly when software automates the Satterthwaite approximation.

- Inference for population spread:

⇒ There are indeed inference procedures appropriate for the standard deviations of Normal populations.

↳ F test for comparing the spread of two Normal populations.

Unlike t procedures, the F-test & other procedures for standard deviations are extremely sensitive to non-Normal distributions.

↳ It's difficult to tell if F-values are evidence of unequal population spreads, or simply evidence that populations are not Normal.

No single measure is adequate to describe the spread of a skewed distributions.

Chapters 14 and 15 discuss procedures called "analysis of variance" for comparing several means. These procedures are extensions of the pooled t test and require that the populations have a common standard deviation. Fortunately, like the t test, analysis of variance comparisons of means are quite robust. (Analysis of variance uses F statistics, but these are not the same as the F statistic for comparing two population standard deviations.) Formal tests for the equality of two standard deviations are not very robust and will often give misleading results. In the words of one distinguished statistician, "To make a preliminary test on variances is rather like putting to sea in a rowing boat to find out whether conditions are sufficiently calm for an ocean liner to leave port!"[30]

- The F-test for equality spread:

⇒ Suppose that we have independent SRSs from two Normal populations, a sample size of n_1 from $N(\mu_1, \sigma_1)$ & a sample size n_2 from $N(\mu_2, \sigma_2)$

⇒ The population means & standard deviations are unknown.

⇒ The hypothesis of equal spread.

$H_0 : \sigma_1 = \sigma_2$.

$H_a : \sigma_1 \neq \sigma_2$.

is tested by a simple statistics, the ratio of sample variance.

The F Statistic and F Distributions

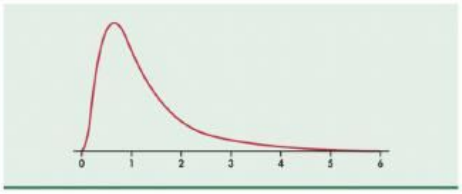
When s_1^2 and s_2^2 are sample variances from independent SRSs of sizes n_1 and n_2 drawn from Normal populations, the ***F* statistic**

$$F = \frac{s_1^2}{s_2^2}$$

has the ***F* distribution** with $n_1 - 1$ and $n_2 - 1$ degrees of freedom when H_0: $\sigma_1 = \sigma_2$ is true.

The ***F* distributions** are a family of distributions with two parameters, the degrees of freedom of the sample variances in the numerator and denominator of the F statistic.*

FIGURE 7.11 The density curve for the $F(9, 10)$ distribution. The F distributions are skewed to the right.



⇒ The numerator degree of freedom is always mentioned first.

↳ Interchanging the degrees of freedom changes the distribution, so order is important

EXAMPLE 7.18 Comparing Healthy and Failed Firms

CASE 7.2 Case 7.2 (page 430) recounts a study that compared the mean ratios of current assets to current liabilities for successful and failed firms. There we used the pooled two-sample t procedures. Because these procedures require equal population standard deviations, it is tempting to first test

$$H_0: \sigma_1 = \sigma_2$$
$$H_a: \sigma_1 \neq \sigma_2$$

The 68 healthy firms had a standard deviation of $s = 0.6393$, while the 33 failed firms had a standard deviation of $s = 0.4811$. The F test statistic is therefore

$$F = \frac{\text{larger } s^2}{\text{smaller } s^2} = \frac{0.6393^2}{0.4811^2} = 1.77$$

We compare the calculated value $F = 1.77$ with critical points for the $F(67, 32)$ distribution. One set of entries in Table E that is close to this distribution is the $F(60, 30)$ distribution. Our observed value $F = 1.77$ falls between the critical values $F^* = 1.74$ and $F^* = 1.94$, corresponding to tail areas 0.05 and 0.025. The two-sided P-value therefore lies between 0.10 and 0.05. The standard deviations are significantly different at the 10% level but not at the 5% level.

In most significance tests, we insist on quite strong evidence before we abandon a null hypothesis. That's because we are trying to convince ourselves and others that the effect described by the alternative hypothesis really is present in the population. In this example, however, the null hypothesis H_0: $\sigma_1 = \sigma_2$ is a requirement for using the pooled t to compare two means. Evidence much weaker than the usual significance levels of 5% or 1% should lead you to question H_0 and abandon the pooled t for the more general t that we have recommended.

- The power of the two-sample t-test :-

⇒ The exact power calculation involves a new distribution, the noncentral - t- distribution.

↳ This calculation is not practical by hand but is easy with software that calculates probabilities.

⇒ We consider only the common case where null hypothesis is "no difference"
$\mu_1 - \mu_2 = 0$.

↳ We illustrate the calculation for pooled two sample-t-test.

↳ A simple modification is needed when we do not pool.

⇒ The unknown parameters in pooled t settings are μ_1, μ_2 and a single σ.

⇒ To find the power, follow these steps.

Step 1. Specify three quantities :

(a) an alternative value for $\mu_1 - \mu_2$ that you consider important to detect;
(b) the sample sizes, n_1 and n_2.
(c) a fixed significance level α, often $\alpha = 0.05$; and
(d) an estimate of the standard deviation σ from a pilot study or previous study under similar conditions.

Step 2: Find the degrees of freedom $df = n_1 + n_2 - 2$, and the value of t^* that will lead to rejecting H_0 at your chosen level α.

Step 3: Calculate the noncentrality parameter.

$$\delta = \frac{|\mu_1 - \mu_2|}{\sigma\sqrt{\frac{1}{n_1} - \frac{1}{n_2}}}.$$

Step 4:

⇒ The power is the probability that a noncentral t random variable with degrees of freedom df and the noncentrality parameter δ will be less than t^*.

If we assume equal sizes, we need to guess a common population standard deviation and then combine them to get an estimate of the standard error.

$$\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

⇒ This guess is then used in denominator.

⇒ Use the conservative value, the smaller of n_1-1, and n_2-1, for degrees of freedom.

EXAMPLE 7.19 Healthy versus Failed Firms

CASE 7.2

In Case 7.2 we compared the ratio of current assets to current liabilities for 68 successful and 33 failed firms. Using the pooled two-sample procedure, the difference was statistically significant ($t = 7.71$, df $= 99$, $P < 0.001$). Now that this study is several years old, we are planning a similar study to determine if the finding continues to hold.

Should our new sample have similar numbers of firms? Or could we save resources by using smaller samples and still be able to declare that the successful and failed firms are different? To answer this question, we do a power calculation.

Step 1 We want to be able to detect a difference in the means that is about the same as the value that we observed in our previous study. So in our calculations we will use $\mu_1 - \mu_2 = 1.15$. We are willing to assume that the standard deviations will be about the same as in the earlier study, so we take the standard deviation for each of the two groups of firms to be the pooled value from our previous study, $\sigma = 0.5928$.

We need only two pieces of additional information: a significance level α and the sample sizes n_1 and n_2. For the first, we will choose the standard value $\alpha = 0.05$. For the sample sizes we want to try several different values. Let's start with $n_1 = 15$ and $n_2 = 15$.

Step 2 The degrees of freedom are $n_1 + n_2 - 2 = 28$. The critical value is $t^* = 2.048$, the value from Table D for a two-sided $\alpha = 0.05$ significance test based on 28 degrees of freedom.

Step 3 The noncentrality parameter is

$$\delta = \frac{1.15}{0.5928\sqrt{\frac{1}{15} + \frac{1}{15}}} = \frac{1.15}{0.2165} = 5.31$$

Step 4 Software gives the power as 0.99922, over 99.9%. The Normal approximation is very accurate:

$$P(Z > t^* - \delta) = P(Z > -3.262) = 0.99945$$

If we repeat the calculation with $n_1 = 10$ and $n_2 = 10$, we still have power greater than 0.98. A difference of means as large as that found in the earlier study can be detected with quite small samples.

A different, and perhaps more important, issue is the margin of error for the estimated difference. For $n_1 = 15$ and $n_2 = 15$, it is

$$t^* \times 0.5928\sqrt{\frac{1}{15} + \frac{1}{15}} = 2.048 \times 5.31 = 0.44$$

Repeating the calculation for $n_1 = 10$ and $n_2 = 10$ gives 0.56. If we want to reduce the margin of error for the difference in the means of asset/liability ratios, we will need larger sample sizes.